ASSIGNMENT 1: HYPOTHESIS TESTING

BAN 100: Statistics for Analytics

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Defining the Hypothesis

Given the data, I've defined with 3 scenarios for Hypothesis (Ho) and Alternative Hypothesis (Ha).

Scenario 1

Hypothesis (Ho): Smoking affects Infant weight.
Alternative Hypothesis (Ha): Smoking does not affect Infant weight.

Scenario 2

Hypothesis (Ho): Male infants weigh more than Female infants.
Alternative Hypothesis (Ha): Male infants do not weigh more than Female infants.

Scenario 3

Hypothesis (Ho): Infant weight is affected if the mother is married.
Alternative Hypothesis (Ha): Infant weight is not affected if the mother is married.

Confidence Interval

Since it is not defined in the question, we assume the confidence interval (**α**) is 0.05

Scenario 1

Hypothesis (Ho): Smoking affects Infant weight.

Alternative Hypothesis (Ha): Smoking does not affect Infant weight.

The TTEST Procedure

Variable: Weight (Weight)

MomSmoke	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		43467	3402.3	558.0	2.6766	240.0	6350.0
1		6533	3160.9	576.8	7.1358	312.0	5245.0
Diff (1-2)	Pooled		241.5	560.5	7.4376		
Diff (1-2)	Satterthwaite		241.5		7.6213		

MomSmoke	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3402.3	3397.1	3407.6	558.0	554.3	561.8
1		3160.9	3146.9	3174.8	576.8	567.0	586.8
Diff (1-2)	Pooled	241.5	226.9	256.0	560.5	557.1	564.0
Diff (1-2)	Satterthwaite	241.5	226.5	256.4			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	32.46	<.0001
Satterthwaite	Unequal	8474.1	31.68	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	6532	43466	1.07	0.0004

Inference:

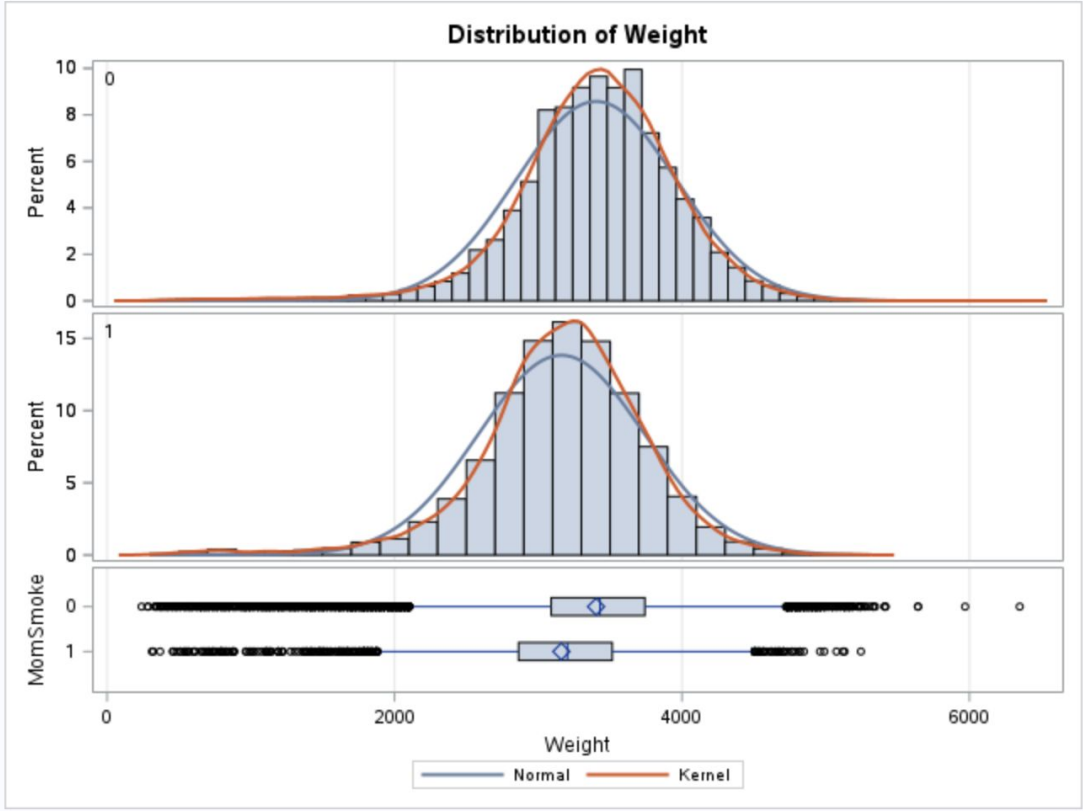
```
proc ttest data=work.import;
class MomSmoke;
var weight;
run;
```

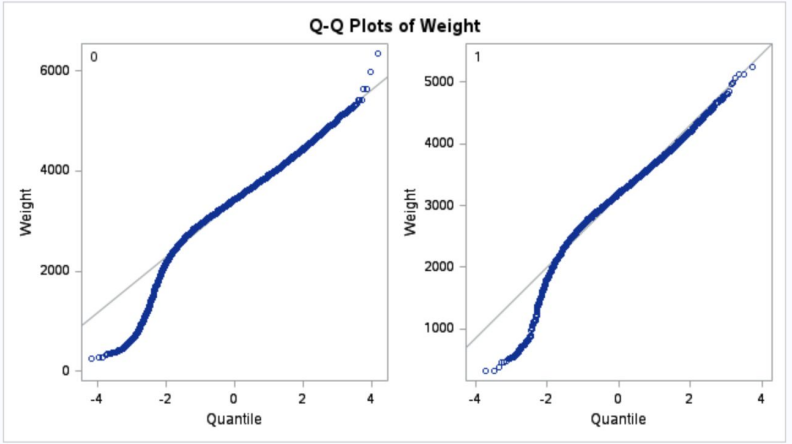
- The 2 variables used in this test are MomSmoke and Weight.
- From the proc ttest, we see that SAS has produced 4 tables.
- In the 4th table called ‘Equality of Variances’, we notice that P value < **α** (0.05)**, so we know that SAS has performed a Satterwaite approximation, which does not assume equal variances.**
- In the 3rd table, looking at the Satterwaite row, we notice that P value < **α** (0.05)**, so we reject the Ho.**

Conclusion: We conclude in favour of the Ha that there is significant evidence that Smoking does not affect Infant weight.

Scenario 1 - Charts

Here are the charts presented by SAS for scenario 1.





Scenario 2

Hypothesis (Ho): Male infants weigh more than Female infants.
Alternative Hypothesis (Ha): Male infants do not weigh more than Female infants.

The TTEST Procedure

Variable: Weight (Weight)

Boy	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		24208	3310.6	547.7	3.5204	240.0	6350.0
1		25792	3427.3	577.7	3.5970	284.0	5970.0
Diff (1-2)	Pooled		-116.7	563.4	5.0416		
Diff (1-2)	Satterthwaite		-116.7		5.0331		

Boy	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3310.6	3303.7	3317.5	547.7	542.9	552.7
1		3427.3	3420.2	3434.3	577.7	572.7	582.7
Diff (1-2)	Pooled	-116.7	-126.6	-106.8	563.4	559.9	566.9
Diff (1-2)	Satterthwaite	-116.7	-126.6	-106.8			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	-23.15	<.0001
Satterthwaite	Unequal	49993	-23.18	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	25791	24207	1.11	<.0001

Inference:

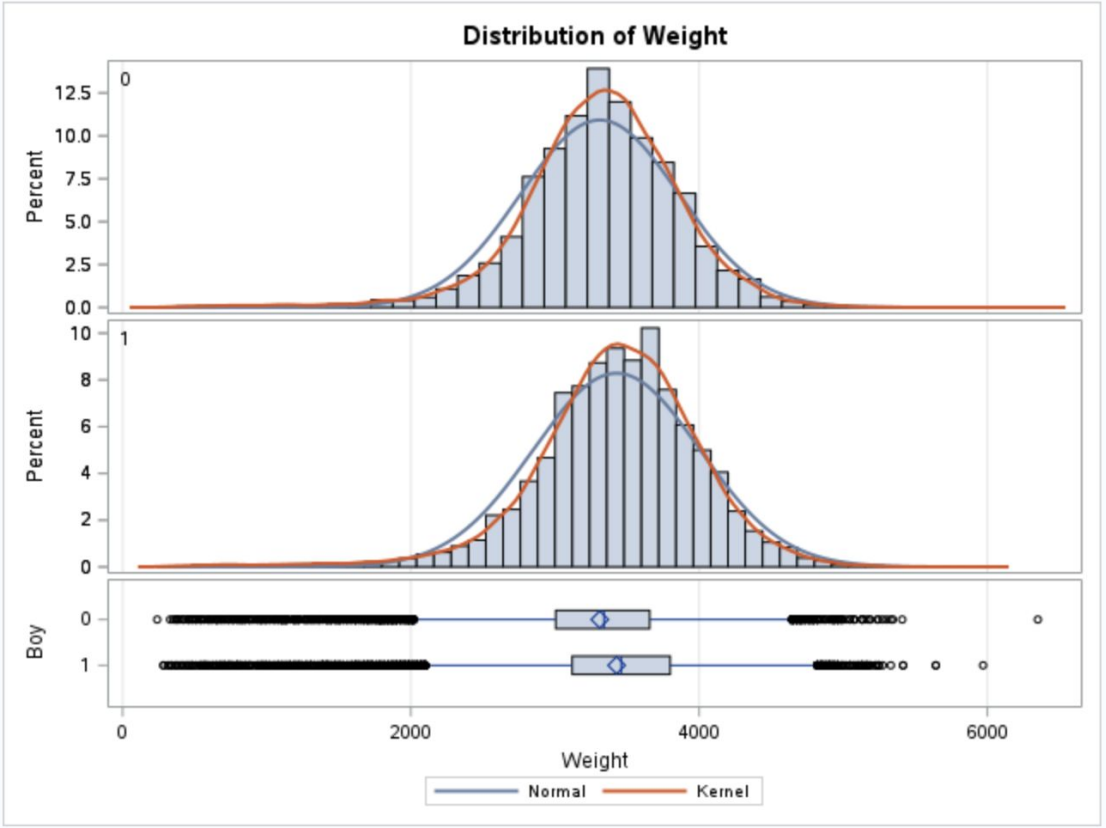
```
proc ttest data=work.import;
class Boy;
var weight;
run;
```

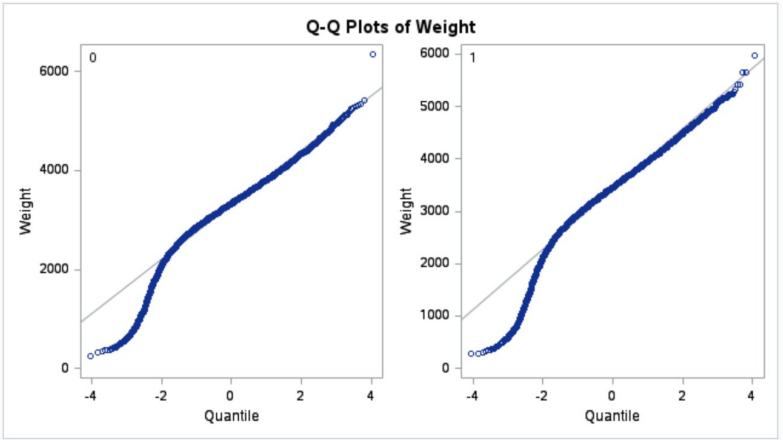
- The 2 variables used in this test are Boy and Weight.
- From the proc ttest, we see that SAS has produced 4 tables.
- In the 4th table called ‘Equality of Variances’, we notice that P value < **α** (0.05)**, so we know that SAS has performed a Satterwaite approximation, which does not assume equal variances.**
- In the 3rd table, looking at the Satterwaite row, we notice that P value < **α** (0.05)**, so we reject the Ho.**

Conclusion: We conclude in favour of the Ha that there is significant evidence that Male infants do not weigh more than Female infants.

Scenario 2 - Charts

Here are the charts presented by SAS for scenario 2.





Scenario 3

Hypothesis (Ho): Infant weight is affected if the mother is married.
Alternative Hypothesis (Ha): Infant weight is not affected if the mother is married.

The TTEST Procedure

Variable: Weight (Weight)

Married	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
0		14369	3234.4	579.0	4.8302	284.0	6350.0
1		35631	3425.7	551.8	2.9231	240.0	5970.0
Diff (1-2)	Pooled		-191.3	559.7	5.5315		
Diff (1-2)	Satterthwaite		-191.3		5.6459		

Married	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
0		3234.4	3225.0	3243.9	579.0	572.4	585.8
1		3425.7	3420.0	3431.5	551.8	547.8	555.9
Diff (1-2)	Pooled	-191.3	-202.1	-180.5	559.7	556.3	563.2
Diff (1-2)	Satterthwaite	-191.3	-202.4	-180.2			

Method	Variances	DF	t Value	Pr > |t|
Pooled	Equal	49998	-34.58	<.0001
Satterthwaite	Unequal	25443	-33.88	<.0001

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	14368	35630	1.10	<.0001

Inference:

```
proc ttest data=work.import;
class Married;
var weight;
run;
```

- The 2 variables used in this test are Boy and Weight.
- From the proc ttest, we see that SAS has produced 4 tables.
- In the 4th table called ‘Equality of Variances’, we notice that P value < **α** (0.05)**, so we know that SAS has performed a Satterwaite approximation, which does not assume equal variances.**
- In the 3rd table, looking at the Satterwaite row, we notice that P value < **α** (0.05)**, so we reject the Ho.**

Conclusion: We conclude in favour of the Ha that there is significant evidence that Infant weight is not affected if the mother is married.

Scenario 3 - Charts

Here are the charts presented by SAS for scenario 3.

